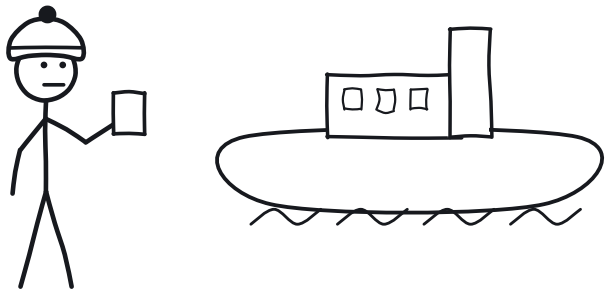
Who Sank the Carbon?

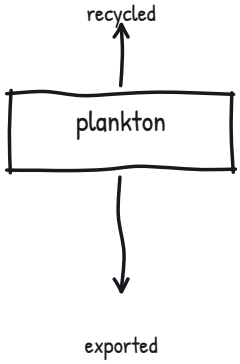
From "Specific eukaryotic plankton are good predictors of net community production in the Western Antarctic Peninsula" by Yajuan Lin, Nicolas Cassar, Adrian Marchetti, Carly Moreno, Hugh Ducklow, Zuchuan Li (2017)

The pump

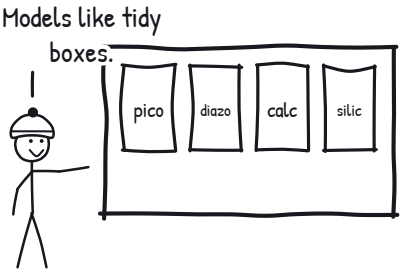
Net community production, NCP, is gross primary production minus community respiration.



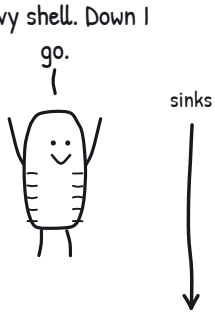
What the plankton make is either recycled or exported to depth, depending on their size and density.



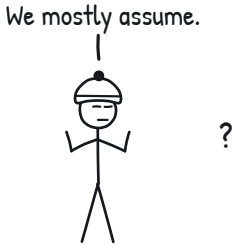
Export is modelled as a function of plankton functional types: picoautotrophs, diazotrophs, calcifiers and silicifiers such as diatoms.



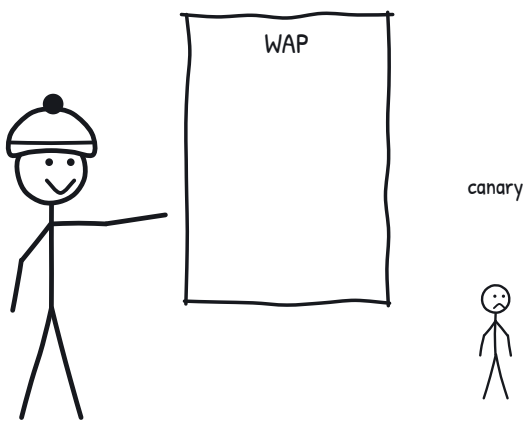
Most diatoms are believed to be efficient exporters because of their large size and heavy silica shells.



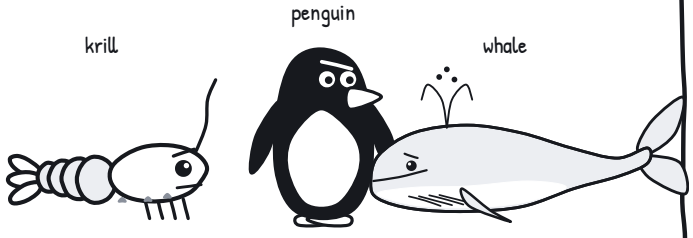
But there is little understanding of how biodiversity influences the carbon pump beyond broad paradigms such as diatoms contributing disproportionally to export.



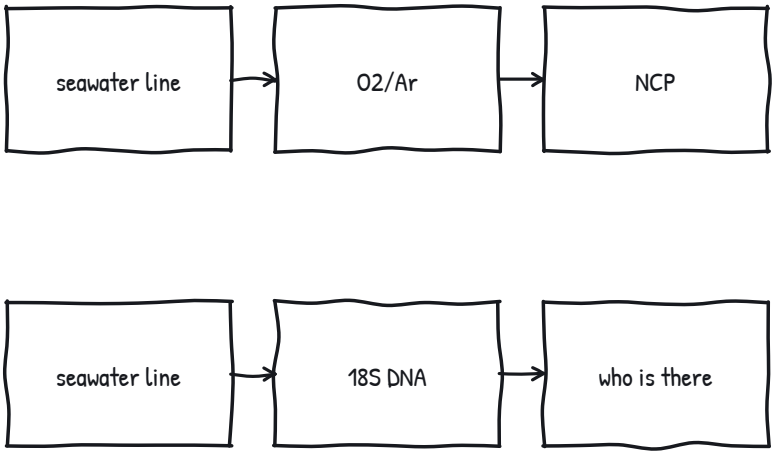
The Western Antarctic Peninsula may be the canary in a coal mine of the Southern Ocean.



Production at the base of the food chain, and upper levels such as krill, fish, penguins, seals and whales, are likely to be affected by rapid warming and sea ice loss.



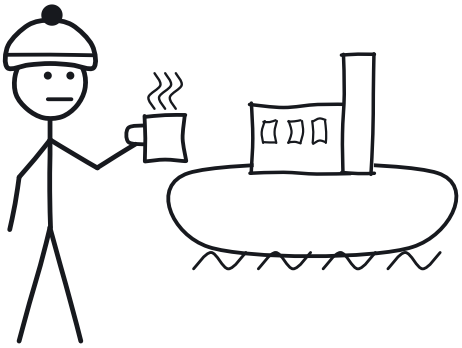
The study combines underway O2/Ar, which estimates net community production, with 18S ribosomal DNA sequencing.



Oxygen stays in the mixed layer 8 to 19 days, 11 on average, within the 1 week to 1 month it takes the plankton community to change. So the two can be compared.



Samples were collected on the annual Palmer LTER cruise, January to early February 2014, aboard the R/V Laurence M. Gould.



Seawater from the ship's supply line was gently filtered through 0.2 μm filters for DNA, while temperature, salinity, oxygen, fluorescence and NCP were measured underway from the same clean flow-through line.



temperature, salinity, oxygen, fluorescence, NCP

DNA was extracted from 21 stations and the V4 region of the 18S gene was sequenced.





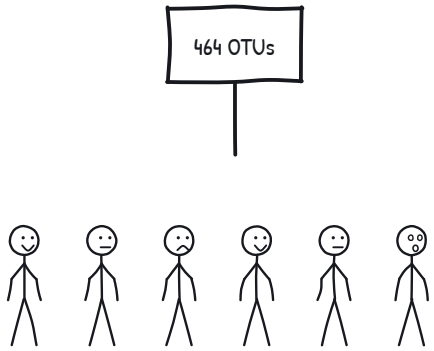
To account for biomass, NCP was normalised to particulate organic carbon, POC. That may indicate how fast the carbon inventory sinks and turns over.

Per unit of me, please.

NCP / POC



At a 97% DNA similarity threshold, roughly species to genus, there were 464 operational taxonomic units, OTUs.



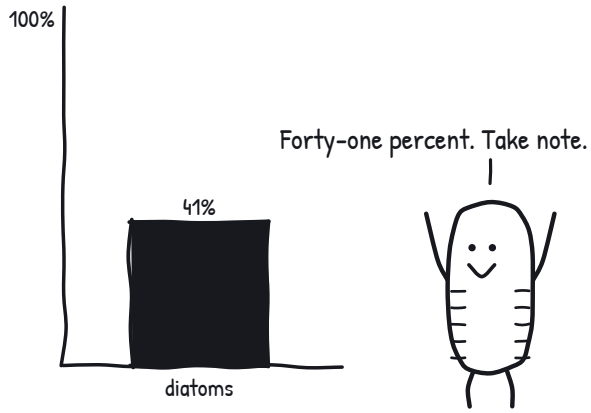
Checks: NCP from O2/Ar correlates highly with NCP from seasonal DIC drawdown (r2 = 0.83). The estimated change in POC averaged 18% of NCP, though that used only 4 comparisons because of cloud cover.



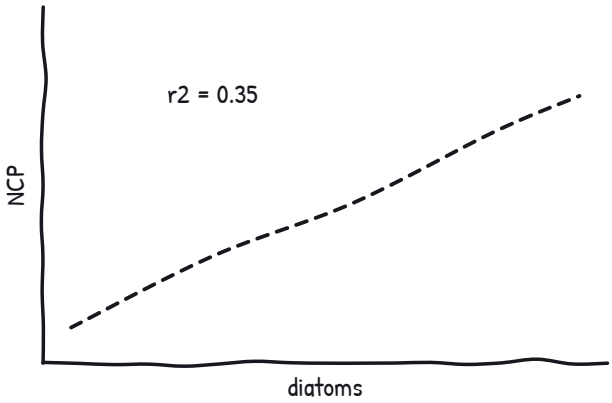


The suspects

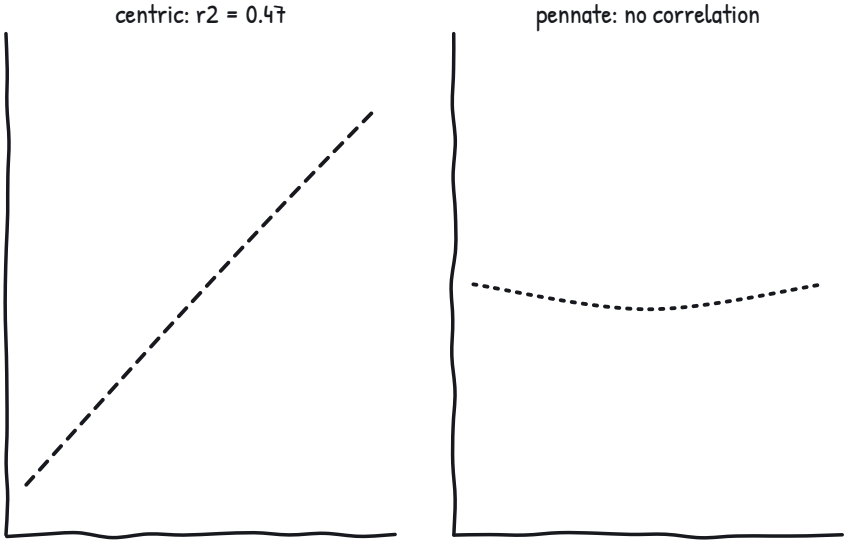
Diatoms dominated the community, an average of 41% of all reads.



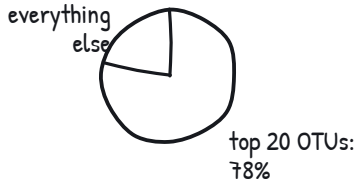
Diatom abundance correlated positively with NCP (r2 = 0.35, p = 0.005), but not significantly with NCP/POC.



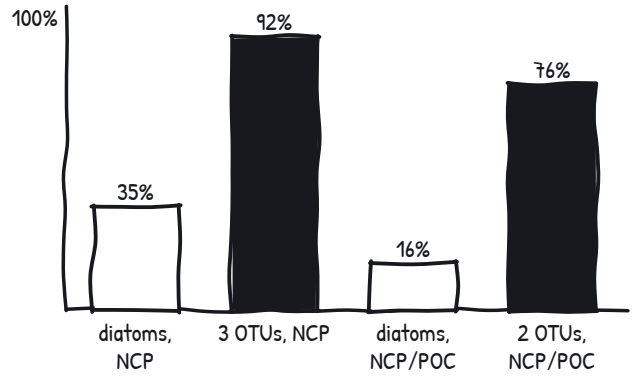
Centric diatoms correlated with NCP (r2 = 0.47, p = 0.0006). Surprisingly, pennate diatoms were not significantly correlated with NCP or NCP/POC.



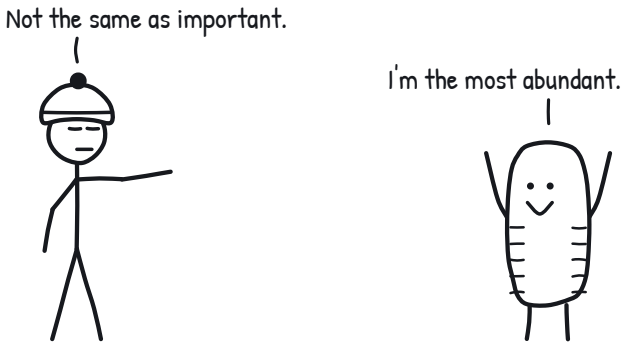
The top 20 OTUs, 4.3% of the diversity, made up 78% of all sequence reads.



Three OTUs explained 92% of the variance in NCP and two OTUs 76% of the variance in NCP/POC. Bulk diatoms explained only 35% and 16%.



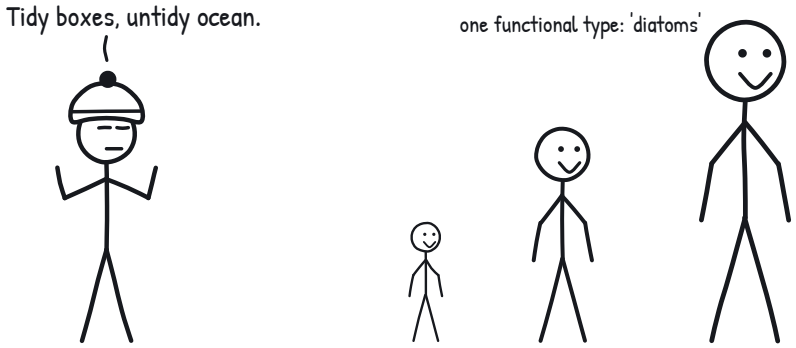
The most dominant OTU, the pennate diatom Fragilariopsis, was not a good predictor of NCP ($r^2 = 0.12$, $p > 0.1$).



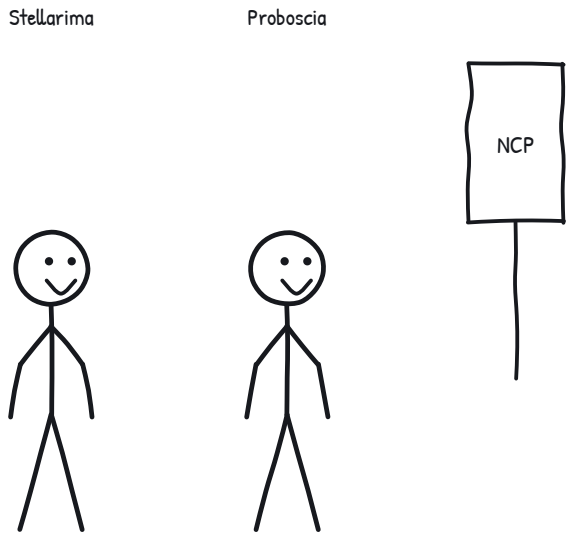
Many Fragilariopsis species are ice algae, so high abundance in some areas may be due to sea ice melt seeding and not new cell growth.



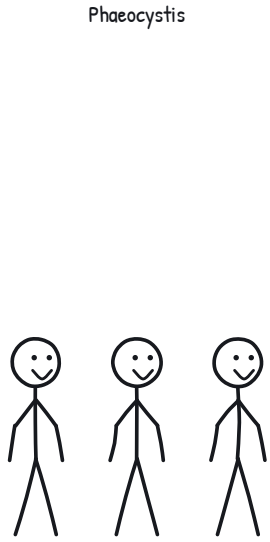
So PFTs may not predict NCP well in some regions, and variability within a PFT may exceed variability between PFTs.



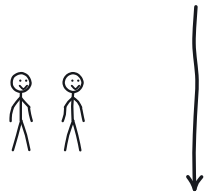
Stellarima in a northern bloom near Palmer Station and Proboscia in the southern ice-edge bloom each correlated with NCP but not NCP/POC: high production, but relatively less efficient export.



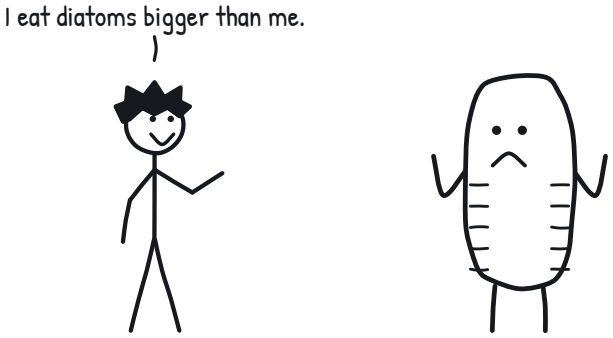
Phaeocystis correlated with NCP/POC, not NCP. It averaged only 9.0% of reads, but at the station with the highest NCP/POC it co-dominated at 34.4% with diatoms at 36.1%.



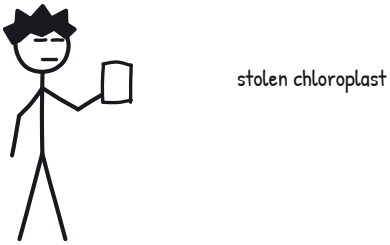
Phaeocystis forms massive aggregates at the colony stage that sink rapidly, and is considered a more efficient exporter of carbon than diatoms.



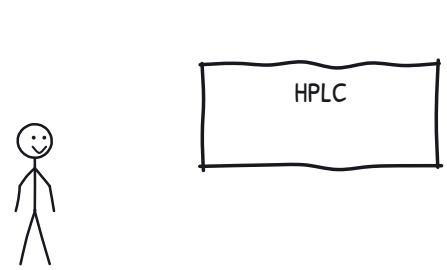
Two dinoflagellate OTUs were key negative contributors to NCP and NCP/POC. Microzooplankton like these are more likely to reduce carbon export than large grazers with fast-sinking faecal pellets.



One dinoflagellate OTU belongs to a kleptoplastic group known to prey on Phaeocystis and keep its chloroplasts.

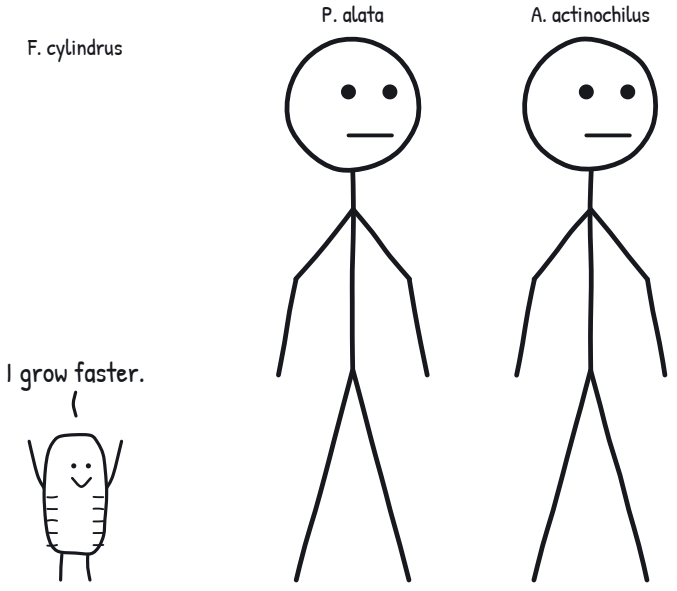


NCP and NCP/POC were also negatively correlated with Rhizaria (r2 = 0.29, p = 0.01 and r2 = 0.20, p = 0.04). Rhizaria are not detected by HPLC pigment analyses because they lack pigments.

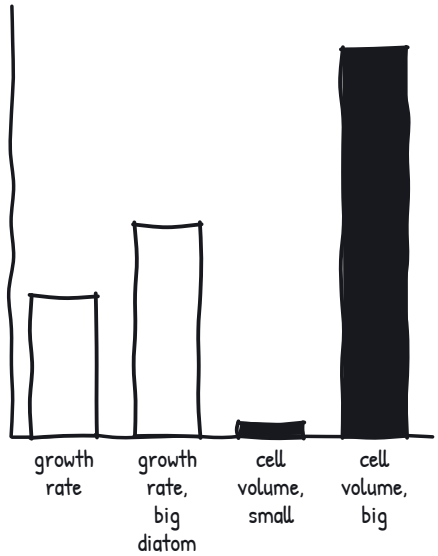


Why these suspects?

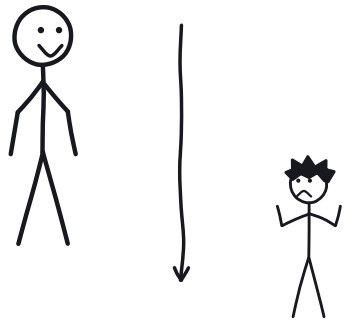
Cultured isolates showed F. cylindrus has a significantly higher maximum growth rate ($P < 0.05$) than Proboscia alata and Actinocyclus actinochilus, yet it does not correlate with NCP.



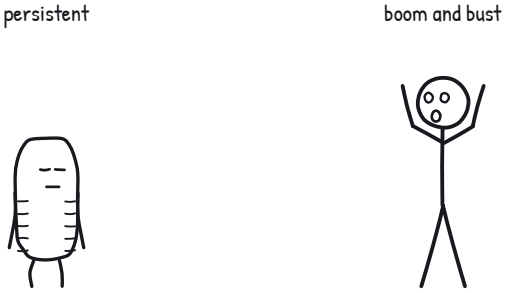
Growth rates of the three diatoms varied by less than 2-fold, but the biovolume of P. alata and A. actinochilus is more than two orders of magnitude larger.



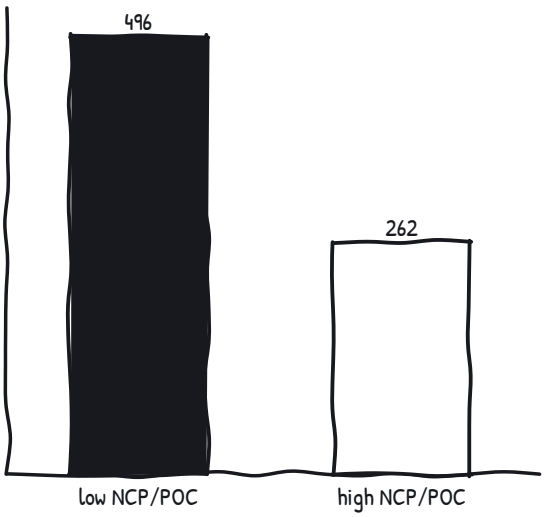
Large size could directly raise sinking rates and export, or uncouple cells from slow-growing grazers.



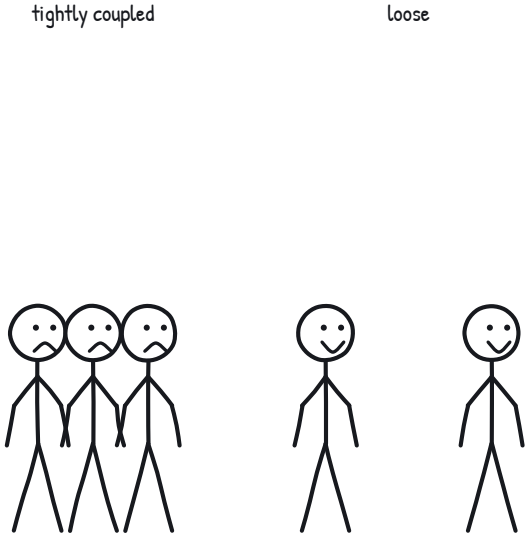
Persistent species are better at stress tolerance. When conditions turn favourable, boom-and-bust species can outpace grazer growth and contribute to more efficient export.



Low NCP/POC communities had almost double the network edges of high NCP/POC communities, 496 against 262, each built from only 11 stations. That suggests tighter coupling.

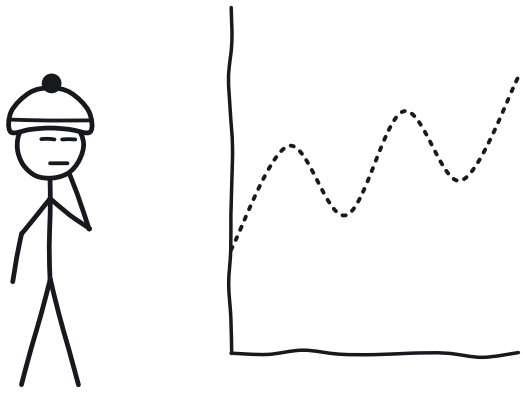


This may support the paradigm that more tightly coupled ecosystems are less efficient exporters of carbon.

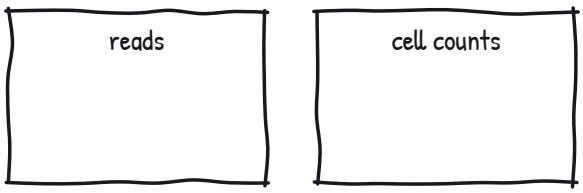


Caveats, and what next

A lack of steady state in the mixed-layer carbon budget may blur some of the relationships.

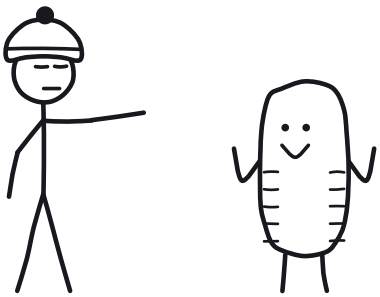


16S and 18S reads differ from taxon-specific cell counts, partly because rDNA copy numbers vary per cell.



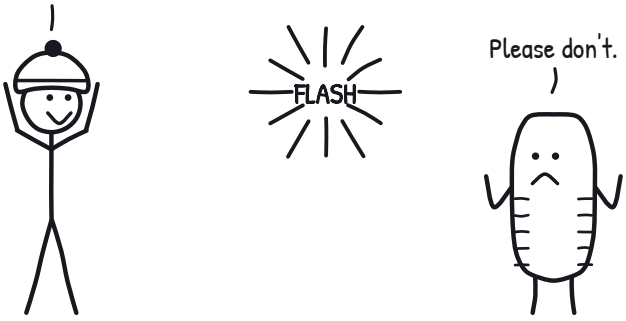
The network and correlation analyses do not necessarily reflect direct causal ecological relationships.

Correlation, Dot.

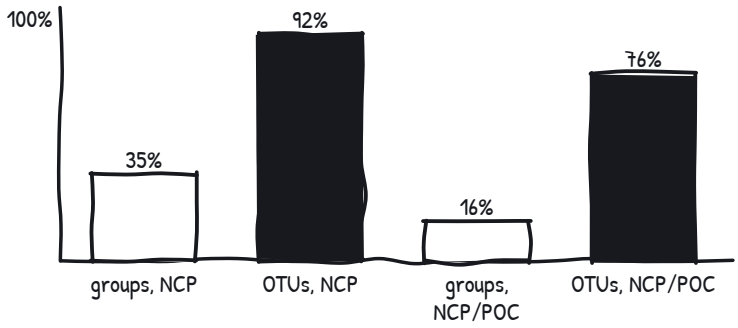


The authors say it will be important to validate the results with in situ growth rates, photophysiology, and microzooplankton growth and grazing rates.

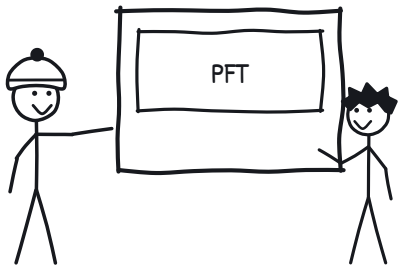
Photophysiology? I know a flash for that.



As few as 2 to 3 key OTUs explain a significant share of the spatial variability in carbon export potential. Using OTUs instead of broad groups raised the variance explained from 35% to 92% for NCP and from 16% to 76% for NCP/POC.

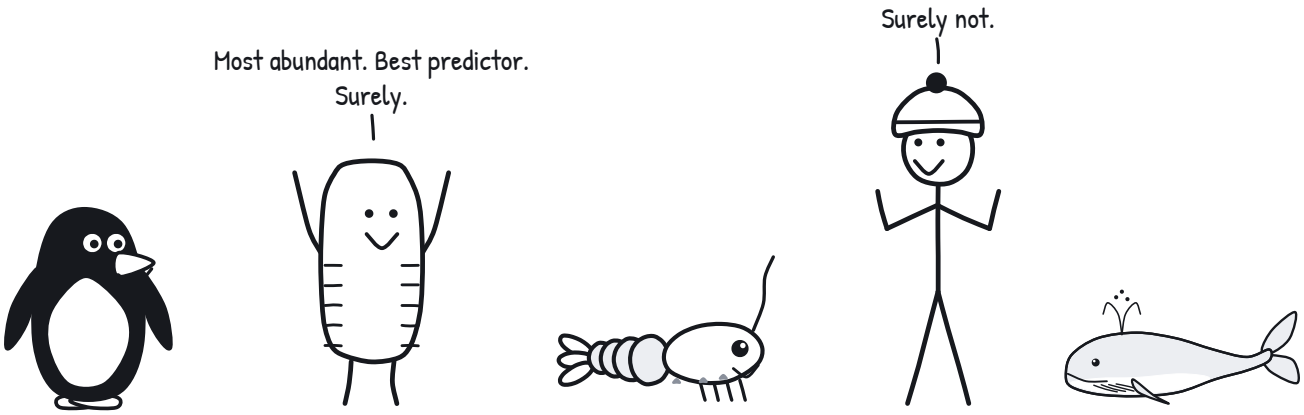


The PFT currency needs adjusting to the process under study. Microzooplankton, dinoflagellates and Rhizaria, emerged as important negative predictors, and functional-group studies should be read with caution.



Most abundant. Best predictor. Surely.

Surely not.



Adapted from "Specific eukaryotic plankton are good predictors of net community production in the Western Antarctic Peninsula" by Yajuan Lin, Nicolas Cassar, Adrian Marchetti, Carly Moreno, Hugh Ducklow, Zuchuan Li (2017), doi 10.1038/s41598-017-14109-1. Licence: CC BY 4.0. Condensed into a comic with invented characters and scenes using the science-comic-pipeline; the paper's authors did not write or endorse this document.

Read the original paper: https://www.nature.com/articles/s41598-017-14109-1

Licence terms: https://creativecommons.org/licenses/by/4.0/